

Statistical Analysis of the Credit Dataset

A Regression Study using R

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Abstract

This report presents a complete statistical analysis of the `Credit` dataset from the *Introduction to Statistical Learning* (ISLR) textbook. The goal is to model credit card *Balance* as a function of seven quantitative and four qualitative predictors. The analysis follows a structured workflow: data inspection, exploratory visualization, normality checks, linear regression modeling (simple, multiple, and with qualitative variables), variable selection via backward selection, and a full set of model diagnostics covering all six potential problems discussed in class — non-linearity, correlated errors, heteroscedasticity, outliers, high-leverage points, and collinearity. A quadratic regression model is ultimately selected as the final model, achieving $R^2 = 0.9695$.

Contents

1	Introduction	3
2	Data Description	3
2.1	Variable Classification	3
2.2	Summary Statistics	4
2.3	Qualitative Variable Counts	4
3	Exploratory Data Analysis	4
3.1	Boxplots of Quantitative Variables	4
3.2	Boxplots of Balance by Qualitative Variables	5
3.3	Scatter Plot Matrix	6
3.4	Correlation Matrix	7
4	Normality Check (Q-Q Plots)	8
5	Simple Linear Regression	9
6	Multiple Linear Regression	10
6.1	Model with All Quantitative Predictors	10
6.2	Collinearity Check — Variance Inflation Factors	11
6.3	Model with Qualitative Predictors (Rating Dropped)	11
6.4	Interaction Term Test	12
6.5	Variable Selection — Backward Selection by p-values	12
7	Model Diagnostics	13
7.1	Residuals vs Fitted (Linear Model)	13
7.2	Non-Linearity: Residuals vs Each Predictor	14
7.3	Remedy: Adding Quadratic Terms	15

7.4	Correlation of Error Terms — Durbin-Watson Test	16
7.5	Non-Constant Variance of Error Terms	16
7.6	Outliers — Studentized Residuals	17
7.7	High-Leverage Points	18
7.8	Collinearity (Re-Check on Final Model)	19
8	Summary of Findings	19
8.1	Final Model	19
8.2	Model Progression	19
8.3	Six Problems — Summary	20
8.4	Conclusion	20
	References	20

1 Introduction

The `Credit` dataset contains information on 400 individuals and their credit card balances. It is taken from the `ISLR` package in R, which accompanies the textbook *An Introduction to Statistical Learning* by James, Witten, Hastie, and Tibshirani.

The objective of this study is to build and validate a linear regression model predicting credit card *Balance* based on both quantitative (income, credit limit, age, etc.) and qualitative predictors (gender, student status, marital status, ethnicity). The analysis also systematically checks all six potential problems in linear regression discussed in class:

1. Non-linearity of response-predictor relationships
2. Correlation of error terms
3. Non-constant variance of error terms (heteroscedasticity)
4. Outliers
5. High-leverage points
6. Collinearity

2 Data Description

The dataset has **400 observations** and **11 variables**: 7 quantitative and 4 qualitative. The response variable is *Balance* (credit card balance in dollars). There are no missing values.

R Code:

```
library(ISLR)
data(Credit)
Credit = Credit[, -1]      # drop the ID column
dim(Credit)
str(Credit)
summary(Credit)
colSums(is.na(Credit))
```

2.1 Variable Classification

Table 1: Classification of Variables in the Credit Dataset

Variable	Type	Description
Income	Quantitative	Income (in \$1,000s)
Limit	Quantitative	Credit limit
Rating	Quantitative	Credit rating score
Cards	Quantitative	Number of credit cards
Age	Quantitative	Age in years
Education	Quantitative	Years of education
Balance	Quantitative	Credit card balance (response)
Gender	Qualitative	Male / Female
Student	Qualitative	Yes / No
Married	Qualitative	Yes / No
Ethnicity	Qualitative	African American / Asian / Caucasian

2.2 Summary Statistics

Table 2: Summary Statistics for Quantitative Variables

Statistic	Income	Limit	Rating	Cards	Age	Education	Balance
Minimum	10.35	855	93	1	23	5	0
1st Qu.	21.01	3088	247.2	2	41.75	11	68.75
Median	33.12	4622	344	3	56	14	459.5
Mean	45.22	4736	354.9	2.958	55.67	13.45	520.01
3rd Qu.	57.47	5873	437.2	4	70	16	863
Maximum	186.63	13913	982	9	98	20	1999
Std. Dev.	35.24	2308.2	154.72	1.37	17.25	3.13	459.76

Inferences:

- *Income*, *Limit*, and *Rating* are all right-skewed (mean > median).
- *Balance* has a minimum of 0, suggesting a large group of customers who carry no balance at all.
- For *Balance*, the mean (520.01) is close to its SD (459.76), indicating very high variability.
- *Age* and *Education* are fairly symmetric around their means.

2.3 Qualitative Variable Counts

- **Gender:** Male = 193, Female = 207
- **Student:** No = 360, Yes = 40 (heavily imbalanced)
- **Married:** No = 155, Yes = 245
- **Ethnicity:** African American = 99, Asian = 102, Caucasian = 199

3 Exploratory Data Analysis

3.1 Boxplots of Quantitative Variables

R Code:

```
par(mfrow = c(2, 4))
boxplot(Credit$Income, col = "skyblue", outcol = "red", outpch = 19,
        main = "Boxplot for Income", ylab = "Income (in $1000)")
boxplot(Credit$Limit, col = "orange", outcol = "red", outpch = 19,
        main = "Boxplot for Limit", ylab = "Credit Limit")
boxplot(Credit$Rating, col = "lightgreen", outcol = "red", outpch = 19,
        main = "Boxplot for Rating", ylab = "Credit Rating")
boxplot(Credit$Cards, col = "pink", outcol = "red", outpch = 19,
        main = "Boxplot for Cards", ylab = "Number of Cards")
boxplot(Credit$Age, col = "purple", outcol = "red", outpch = 19,
        main = "Boxplot for Age", ylab = "Age (years)")
boxplot(Credit$Education, col = "yellow", outcol = "red", outpch = 19,
        main = "Boxplot for Education", ylab = "Years of Education")
boxplot(Credit$Balance, col = "cyan", outcol = "red", outpch = 19,
        main = "Boxplot for Balance", ylab = "Credit Card Balance")
par(mfrow = c(1, 1))
```

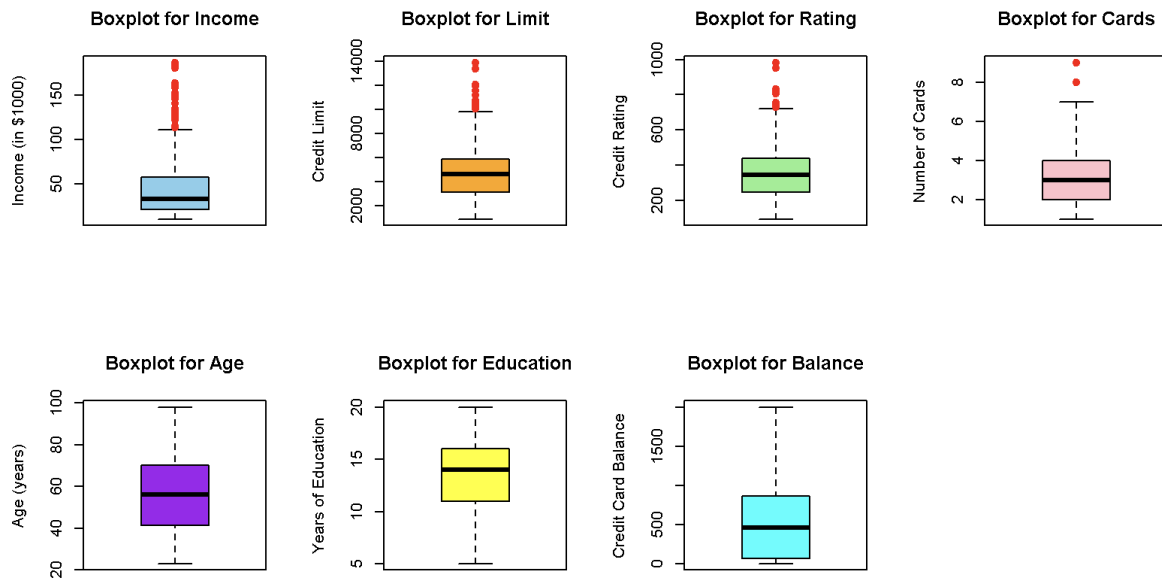


Figure 1: Boxplots of all seven quantitative variables

Inferences:

- *Income*, *Limit*, *Rating*, and *Cards* all show upper outliers (red dots).
- *Age* and *Education* show no outliers and look approximately symmetric.
- *Balance* is right-skewed with a lower whisker nearly touching zero, confirming the zero-balance cluster.

3.2 Boxplots of Balance by Qualitative Variables**R Code:**

```
par(mfrow = c(2, 2))
boxplot(Balance ~ Gender, data = Credit, col = c("skyblue", "pink"),
        outcol = "red", outpch = 19, main = "Balance by Gender")
boxplot(Balance ~ Student, data = Credit, col = c("lightgreen", "orange"),
        outcol = "red", outpch = 19, main = "Balance by Student")
boxplot(Balance ~ Married, data = Credit, col = c("yellow", "purple"),
        outcol = "red", outpch = 19, main = "Balance by Married")
boxplot(Balance ~ Ethnicity, data = Credit, col = c("cyan", "pink", "lightgreen"),
        outcol = "red", outpch = 19, main = "Balance by Ethnicity")
par(mfrow = c(1, 1))
```

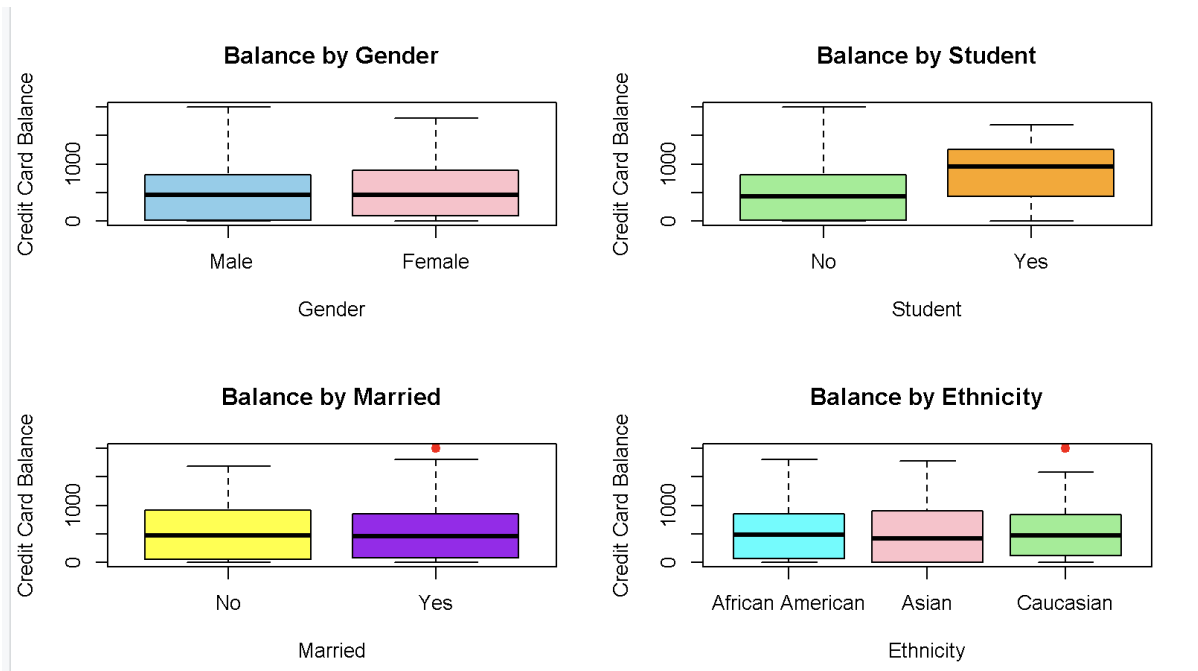


Figure 2: Credit card Balance grouped by qualitative predictors

Inferences:

- **Student** shows the strongest effect — students have roughly double the median balance of non-students.
- **Gender**, **Married**, and **Ethnicity** show nearly identical distributions across groups. These are unlikely to be useful predictors.

3.3 Scatter Plot Matrix**R Code:**

```
quant_data = Credit[, c("Income", "Limit", "Rating", "Cards",
                        "Age", "Education", "Balance")]
pairs(quant_data,
      main = "Scatter Plot Matrix of Quantitative Variables",
      pch = 19, col = "darkgreen", cex = 0.5)
```

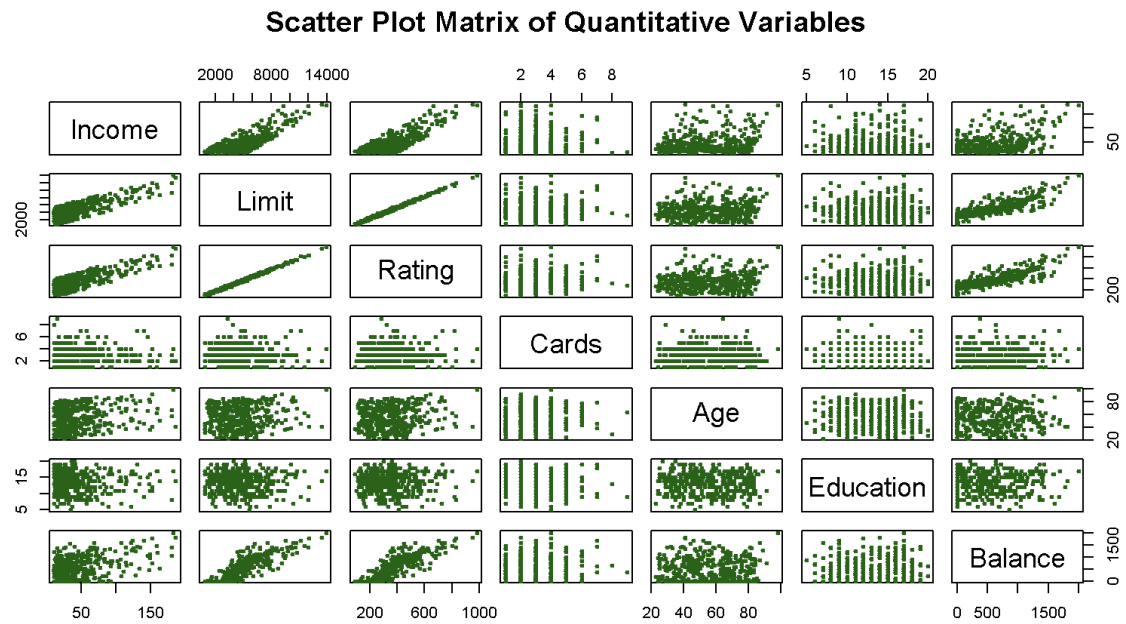


Figure 3: Scatter plot matrix of quantitative variables

Inferences:

- *Limit* and *Rating* form a near-perfect straight line — a strong sign of collinearity.
- *Balance* increases linearly with *Limit* and *Rating*.
- *Balance* vs *Income* is wedge-shaped, hinting at heteroscedasticity.
- *Cards*, *Age*, and *Education* show no clear pattern with *Balance*.

3.4 Correlation Matrix**R Code:**

```
cor_matrix = cor(quant_data)
round(cor_matrix, 3)

library(corrplot)
corrplot(cor_matrix, method = "color", type = "upper",
         addCoef.col = "black", tl.col = "black", tl.srt = 45,
         number.cex = 0.9,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         main = "Correlation Matrix of Quantitative Variables",
         mar = c(0, 0, 2, 0))
```

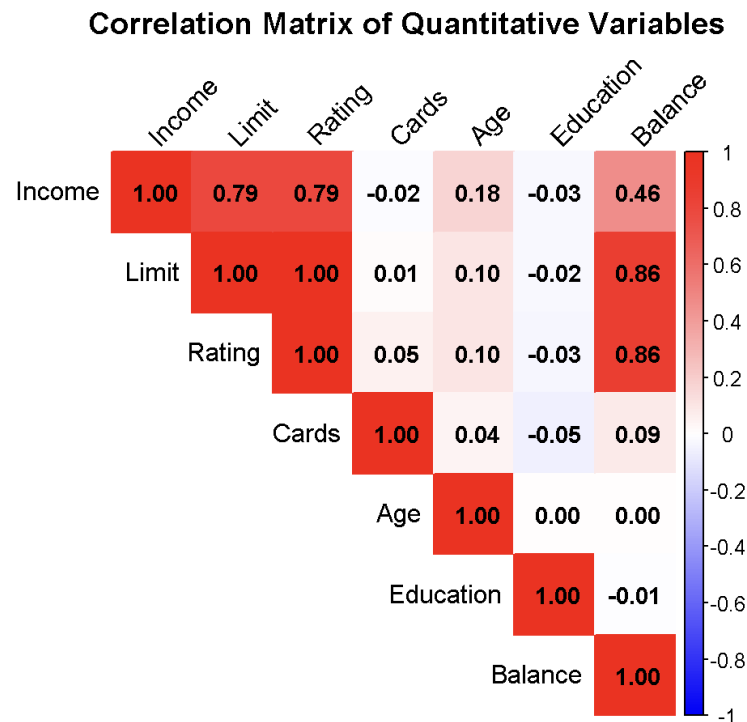


Figure 4: Correlation heatmap of quantitative variables

Inferences:

- **Limit vs Rating:** $r = 1.00$ (rounded) — confirms extreme collinearity.
- **Limit vs Balance** and **Rating vs Balance:** $r = 0.86$ (very strong).
- **Income vs Balance:** $r = 0.46$ (moderate).
- *Cards, Age, and Education* correlate with *Balance* at $r \approx 0$ (essentially no linear relationship).

4 Normality Check (Q-Q Plots)

R Code:

```
par(mfrow = c(2, 2))
for (var_name in c("Balance", "Income", "Limit", "Rating")) {
  my_data = Credit[[var_name]]
  x_ordered = sort(my_data)
  n = length(x_ordered)
  i = 1:n
  p_i = (i - 0.5) / n
  q_i = qnorm(p_i, mean = mean(my_data), sd = sd(my_data))
  plot(q_i, x_ordered, pch = 19, col = "darkgreen",
       main = paste("Normal Q-Q Plot:", var_name),
       xlab = "Theoretical Quantiles",
       ylab = "Sample Order Statistics")
  abline(0, 1, col = "red", lwd = 2)
}
par(mfrow = c(1, 1))
```

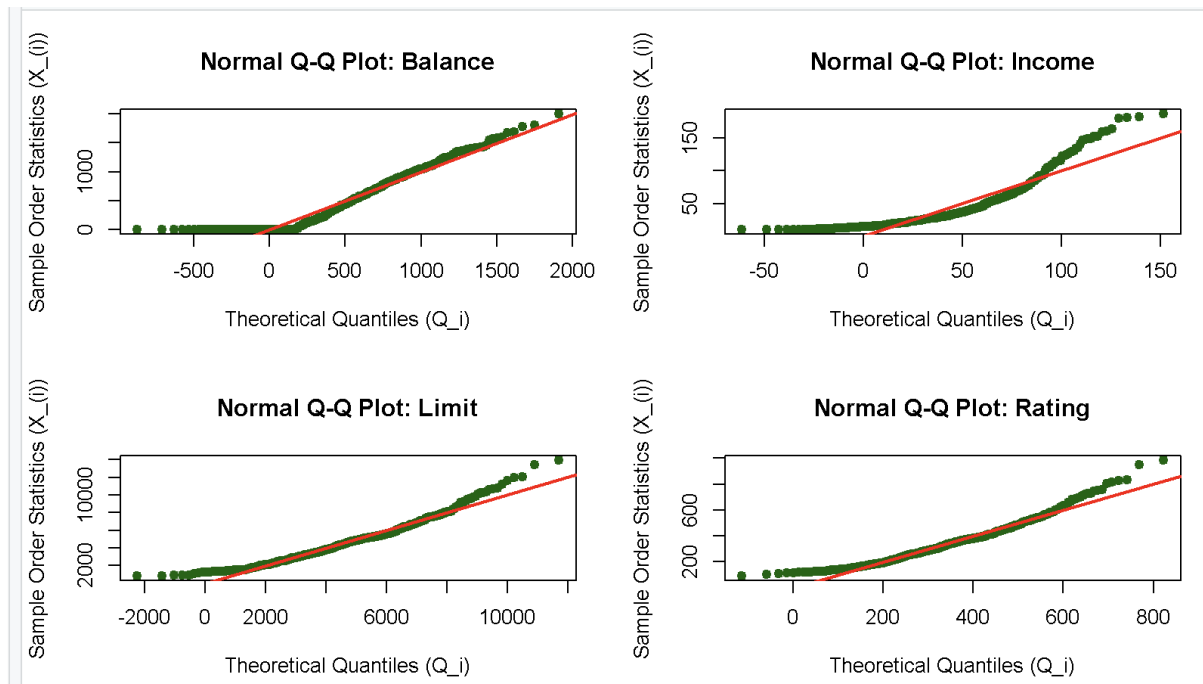



Figure 5: Normal Q-Q plots for key variables

Inferences:

- **Balance** deviates heavily from the line, with a flat segment on the left caused by the zero-balance cluster.
- **Income** shows a clear S-shape, confirming right-skew.
- **Limit** and **Rating** are approximately normal with minor tail deviations.

Note: OLS regression does not require the predictors or the response to be normally distributed; it requires the *residuals* to be approximately normal. This is checked later in the diagnostic phase.

5 Simple Linear Regression

As a baseline, we fit a simple linear regression of *Balance* on *Income*. *Income* was chosen for this role because it is the most interpretable predictor in the dataset and has a moderate correlation with *Balance* ($r = 0.46$), making it a reasonable starting point. The more strongly correlated variables *Limit* and *Rating* ($r = 0.86$ each) were deliberately avoided at this stage, since they are almost perfectly collinear with each other and would be more meaningfully analyzed later in the multiple regression framework. Starting with *Income* also establishes a clear baseline R^2 which we can later compare against the multiple regression models.

R Code:

```
attach(Credit)
lm.fit = lm(Balance ~ Income, data = Credit)
summary(lm.fit)
confint(lm.fit)

plot(Income, Balance, pch = 19, col = "darkgreen",
     main = "Simple Linear Regression: Balance ~ Income",
     xlab = "Income (in $1000)", ylab = "Credit Card Balance")
abline(lm.fit, col = "red", lwd = 3)
```

$$\widehat{\text{Balance}} = 246.51 + 6.05 \cdot \text{Income}$$

Table 3: Simple Linear Regression: Balance ~ Income

Coefficient	Estimate	Std. Error	t-value	p-value
(Intercept)	246.515	33.199	7.43	6.9×10^{-13} ***
Income	6.048	0.579	10.44	$< 2 \times 10^{-16}$ ***

Model fit: $R^2 = 0.215$, Adjusted $R^2 = 0.213$, RSE = 407.9, F = 109.

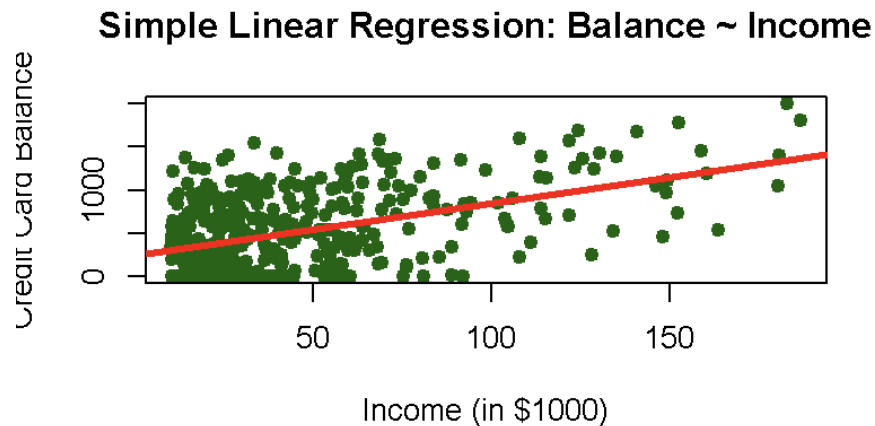


Figure 6: Simple linear regression: Balance vs Income

Inferences:

- *Income* is highly significant, but alone it explains only 21.5% of the variation in *Balance*.
- The wedge-shape of the points around the line suggests heteroscedasticity.
- A multiple regression is clearly needed to capture the remaining variation.

6 Multiple Linear Regression

The Goal of Multiple Regression

After the simple regression gave us $R^2 = 0.215$ (Income alone explained only 21.5% of Balance), we needed a better model. The idea is to use **multiple predictors together** to capture more of what drives Balance.

6.1 Model with All Quantitative Predictors

We now include all six quantitative predictors.

R Code:

```
lm.fit = lm(Balance ~ Income + Limit + Rating + Cards + Age + Education
            ,
            data = Credit)
summary(lm.fit)

library(car)
vif(lm.fit)
```

Table 4: Multiple Regression (quantitative predictors)

Coefficient	Estimate	Std. Error	t-value	p-value
(Intercept)	-477.96	55.07	-8.68	$< 2 \times 10^{-16}$ ***
Income	-7.558	0.382	-19.77	$< 2 \times 10^{-16}$ ***
Limit	0.126	0.053	2.37	0.018 *
Rating	2.063	0.794	2.60	0.0097 **
Cards	11.592	7.067	1.64	0.102
Age	-0.892	0.478	-1.87	0.063 .
Education	1.998	2.600	0.77	0.443

Model fit: $R^2 = 0.878$, Adjusted $R^2 = 0.876$, RSE = 161.6, F = 472.5.

6.2 Collinearity Check — Variance Inflation Factors

Table 5: VIFs with Rating included

Variable	VIF
Income	2.77
Limit	228.85
Rating	230.61
Cards	1.43
Age	1.04
Education	1.01

Inferences:

- Per Lecture-29 slide 20, a VIF above 5 or 10 indicates problematic collinearity. *Limit* and *Rating* have VIFs above 200 — extreme collinearity.
- Income's coefficient has flipped to negative, which is a classic warning sign of collinearity.
- We drop *Rating* (since *Limit* is more directly interpretable) and re-fit the model.

6.3 Model with Qualitative Predictors (Rating Dropped)

We now include all four qualitative predictors. R automatically encodes factor variables using dummy variables.

R Code:

```
lm.fit = lm(Balance ~ Income + Limit + Cards + Age + Education +
            Gender + Student + Married + Ethnicity,
            data = Credit)
summary(lm.fit)
vif(lm.fit)

# Check the dummy encoding used by R
contrasts(Credit$Gender)
contrasts(Credit$Student)
contrasts(Credit$Married)
contrasts(Credit$Ethnicity)
```

Table 6: Multiple Regression (quantitative + qualitative, no Rating)

Coefficient	Estimate	Std. Error	t-value	p-value
(Intercept)	-448.9	33.49	-13.41	$< 2 \times 10^{-16}$ ***
Income	-7.768	0.235	-33.05	$< 2 \times 10^{-16}$ ***
Limit	0.2664	0.0035	75.10	$< 2 \times 10^{-16}$ ***
Cards	23.27	3.642	6.39	4.79×10^{-10} ***
Age	-0.608	0.296	-2.06	0.040 *
Education	-1.385	1.602	-0.86	0.388
Gender (Female)	-10.55	9.97	-1.06	0.291
Student (Yes)	429.3	16.74	25.64	$< 2 \times 10^{-16}$ ***
Married (Yes)	-5.91	10.36	-0.57	0.569
Ethnicity (Asian)	14.72	14.17	1.04	0.299
Ethnicity (Cauc.)	9.89	12.28	0.81	0.421

Model fit: $R^2 = 0.9545$, Adjusted $R^2 = 0.9533$, RSE = 99.34, F = 815.7.

Inferences:

- **Student** is a major predictor — being a student adds about \$429 to expected balance.
- *Gender*, *Married*, *Ethnicity*, and *Education* are all insignificant.
- *Cards* is now significant after controlling for other variables.
- After dropping *Rating*, all VIFs are below 3 — collinearity is fixed.

6.4 Interaction Term Test

Following Lecture-28, we tested the *Income* $\times$ *Student* interaction using an ANOVA comparison.

R Code:

```
lm.fit_int = lm(Balance ~ Income * Student + Limit + Cards + Age +
                Education + Gender + Married + Ethnicity,
                data = Credit)
summary(lm.fit_int)
anova(lm.fit, lm.fit_int)
```

Table 7: ANOVA: with vs without interaction

Model	Res.Df	RSS	F	p-value
Without interaction	389	3,839,044	—	—
With interaction	388	3,830,636	0.85	0.357

Inference: The interaction is **not significant** ($p = 0.357$). The effect of *Student* is a constant shift of about \$429, independent of income. We do not include the interaction in the final model.

6.5 Variable Selection — Backward Selection by p-values

Following Lecture-28 slide 5, we apply **backward selection based on p-values**: starting with all predictors, we drop the variable with the *largest* (least significant) p-value at each step, then re-fit the model and repeat. We stop when all remaining predictors are significant at the 5% level.

R Code:

```
# Start with all predictors
lm.fit = lm(Balance ~ Income + Limit + Cards + Age + Education +
            Gender + Student + Married + Ethnicity,
            data = Credit)
```

```

# Iteratively drop the predictor with the highest p-value
# until all remaining p-values < 0.05

# Step 1: Drop Ethnicity (largest p-values, ~0.30-0.42)
lm.fit = update(lm.fit, . ~ . - Ethnicity)

# Step 2: Drop Married (p = 0.569)
lm.fit = update(lm.fit, . ~ . - Married)

# Step 3: Drop Education (p = 0.388)
lm.fit = update(lm.fit, . ~ . - Education)

# Step 4: Drop Gender (p = 0.291)
lm.fit = update(lm.fit, . ~ . - Gender)

# Final model - all remaining p-values < 0.05
summary(lm.fit)

```

Table 8: Backward Selection Path (Highest p-value Dropped at Each Step)

Step	Variable Dropped	p-value at step
Start	—	—
1	Ethnicity	0.30–0.42 (both levels)
2	Married	0.569
3	Education	0.388
4	Gender	0.291
Stop	(all remaining predictors significant at 5%)	—

The selected model is:

$$\text{Balance} \sim \text{Income} + \text{Limit} + \text{Cards} + \text{Age} + \text{Student}$$

Final linear model fit: $R^2 = 0.9541$, Adjusted $R^2 = 0.9535$, $\text{RSE} = 99.12$, $F = 1638$. In this final model, every remaining predictor has $p < 0.05$.

7 Model Diagnostics

We now test the five-predictor linear model against all six potential regression problems described in Lecture-29.

7.1 Residuals vs Fitted (Linear Model)

Per Lecture-29 slides 4–5, the Residuals vs Fitted plot is used to detect non-linearity and heteroscedasticity. A clear pattern (curve, funnel, or systematic shape) indicates a problem.

R Code:

```

lm.final = lm(Balance ~ Income + Limit + Cards + Age + Student,
              data = Credit)

plot(predict(lm.final), residuals(lm.final),
     pch = 19, col = "darkgreen",
     main = "Residuals vs Fitted Values (Linear Model)",
     xlab = "Fitted Values", ylab = "Residuals")

```

```
abline(h = 0, col = "red", lwd = 2, lty = 2)
lines(lowess(predict(lm.final), residuals(lm.final)),
      col = "blue", lwd = 2)
```

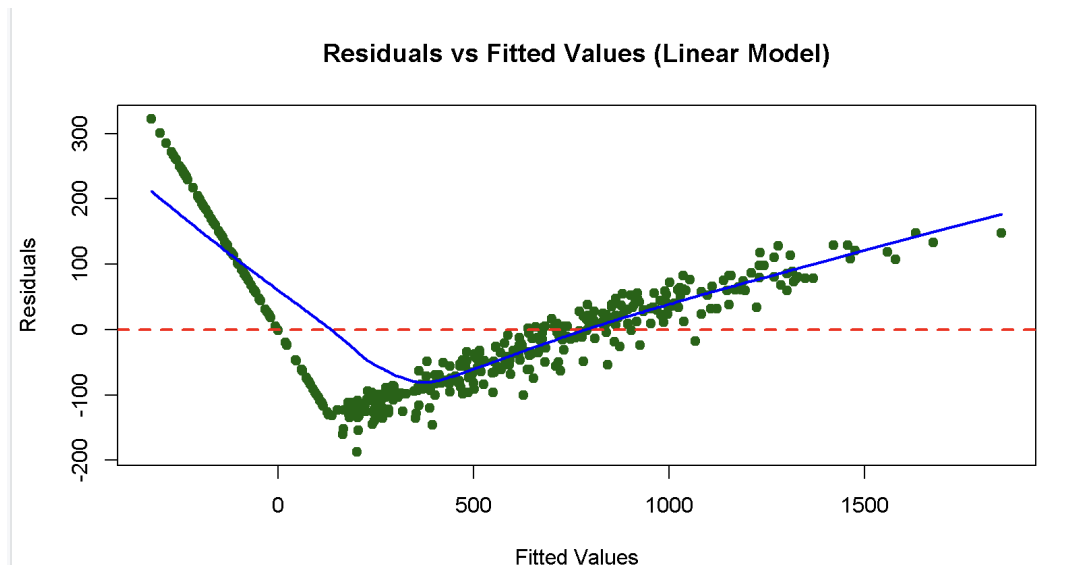


Figure 7: Residuals vs Fitted Values for the linear model

Inference: The residuals show a clear V-shaped / hockey-stick curvature rather than scattering randomly around the zero line. This is a strong signal of **non-linearity** in the model (Lecture-29 slides 4–5). The blue lowess smoother deviates visibly from the red dashed zero line, especially for smaller fitted values, confirming that one or more predictors have a non-linear relationship with *Balance*. A remedy using non-linear transformations will be applied next.

7.2 Non-Linearity: Residuals vs Each Predictor

To identify which predictor is responsible for the non-linearity, we plot residuals against each quantitative predictor separately.

R Code:

```
par(mfrow = c(2, 2))
plot(Credit$Income, residuals(lm.final), pch = 19, col = "darkgreen",
     main = "Residuals vs Income", xlab = "Income", ylab = "Residuals")
abline(h = 0, col = "red", lwd = 2, lty = 2)
plot(Credit$Limit, residuals(lm.final), pch = 19, col = "darkgreen",
     main = "Residuals vs Limit", xlab = "Limit", ylab = "Residuals")
abline(h = 0, col = "red", lwd = 2, lty = 2)
plot(Credit$Cards, residuals(lm.final), pch = 19, col = "darkgreen",
     main = "Residuals vs Cards", xlab = "Cards", ylab = "Residuals")
abline(h = 0, col = "red", lwd = 2, lty = 2)
plot(Credit$Age, residuals(lm.final), pch = 19, col = "darkgreen",
     main = "Residuals vs Age", xlab = "Age", ylab = "Residuals")
abline(h = 0, col = "red", lwd = 2, lty = 2)
par(mfrow = c(1, 1))
```

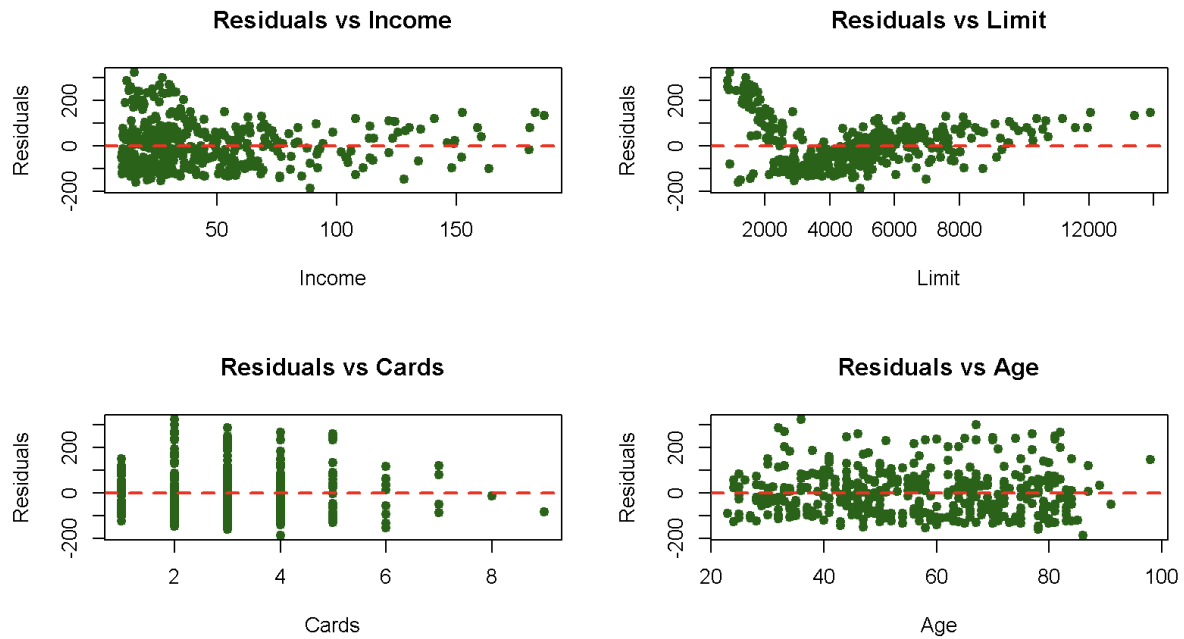


Figure 8: Residuals vs each quantitative predictor

Inferences:

- *Residuals vs Limit* shows a strong curved pattern on the left — *Limit* is the main source of non-linearity.
- *Residuals vs Income* shows a funnel shape, an early sign of heteroscedasticity.
- *Residuals vs Cards* and *Residuals vs Age* look fine.

7.3 Remedy: Adding Quadratic Terms

Based on the pattern above, and following the advice in Lecture-29 slide 6 to use non-linear transformations such as $\log X$, $\sqrt{X}$, or X^2 , we add $I(\text{Income}^2)$ and $I(\text{Limit}^2)$ terms.

R Code:

```
lm.quadratic = lm(Balance ~ Income + I(Income^2) + Limit + I(Limit^2) +
                  Cards + Age + Student,
                  data = Credit)
summary(lm.quadratic)
anova(lm.final, lm.quadratic)
```

Table 9: Quadratic Model Summary

Coefficient	Estimate	Std. Error	t-value	p-value
(Intercept)	-283.9	22.98	-12.36	$< 2 \times 10^{-16}$ ***
Income	-6.436	0.447	-14.40	$< 2 \times 10^{-16}$ ***
$I(\text{Income}^2)$	-0.02047	0.00315	-6.51	2.34×10^{-10} ***
Limit	0.1704	0.0076	22.54	$< 2 \times 10^{-16}$ ***
$I(\text{Limit}^2)$	1.03×10^{-5}	7.52×10^{-7}	13.70	$< 2 \times 10^{-16}$ ***
Cards	22.41	2.968	7.55	3.09×10^{-13} ***
Age	-0.738	0.240	-3.07	0.0023 **
Student (Yes)	440.9	13.56	32.51	$< 2 \times 10^{-16}$ ***

Quadratic model fit: $R^2 = 0.9695$, Adjusted $R^2 = 0.969$, $RSE = 81.01$, $F = 1780$.

An ANOVA test comparing the linear and quadratic models gives $F = 98.93$ with $p\text{-value} < 2.2 \times 10^{-16}$, confirming that the quadratic terms significantly improve the model. We adopt the quadratic model as the final model going forward.

7.4 Correlation of Error Terms — Durbin-Watson Test

As described in Lecture-29 slide 9, the Durbin-Watson test checks whether the error terms are correlated.

R Code:

```
library(car)
durbinWatsonTest(lm.quadratic)
```

- DW statistic = 1.9534
- p-value = 0.606

Inference: DW is very close to 2, and the p-value is large. We **fail to reject** the null hypothesis of no autocorrelation. Errors are independent, which makes sense since *Credit* is cross-sectional data.

7.5 Non-Constant Variance of Error Terms

Following Lecture-29 slides 10–11, the Residuals vs Fitted plot is used to detect heteroscedasticity. A funnel shape or curved pattern indicates non-constant variance.

R Code:

```
plot(predict(lm.quadratic), residuals(lm.quadratic),
     pch = 19, col = "darkgreen",
     main = "Residuals vs Fitted Values (Final Model)",
     xlab = "Fitted Values", ylab = "Residuals")
abline(h = 0, col = "red", lwd = 2, lty = 2)
lines(lowess(predict(lm.quadratic), residuals(lm.quadratic)),
      col = "blue", lwd = 2)
```

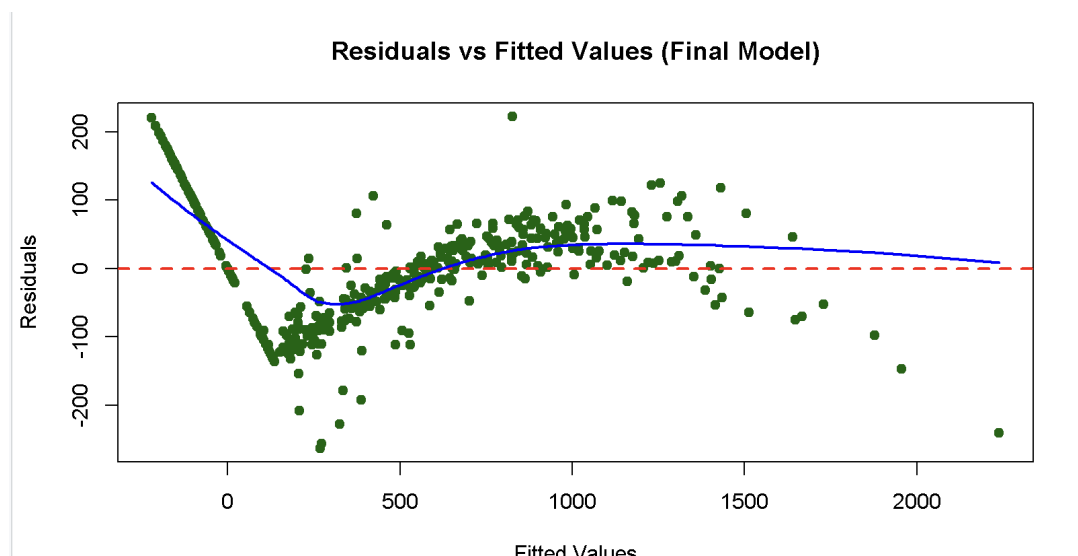


Figure 9: Residuals vs Fitted Values for the Quadratic Model

Inference: Compared to the linear model, the curvature in the residuals has been significantly reduced by the quadratic transformation — the blue lowess line is much flatter around zero. However, the residuals still do not spread uniformly across the range of fitted values: there is larger scatter for small fitted values and the points cluster tightly near zero at the low end. This matches the funnel/curve shapes

shown in Lecture-29 slide 11 and indicates that **heteroscedasticity is still present**. The main cause is the large group of zero-balance observations, which creates high residual variance at low fitted values. As suggested in Lecture-29 slides 12–13, a transformation of the response (such as $\log y$ or $\sqrt{y}$) or weighted least squares could address this issue, but we leave the model untransformed for interpretability.

7.6 Outliers — Studentized Residuals

Per Lecture-29 slide 14, an observation is flagged as an outlier if its studentized residual exceeds 3 in absolute value.

R Code:

```
stud_resid = rstudent(lm.quadratic)
outliers_idx = which(abs(stud_resid) > 3)
Credit[outliers_idx, ]
stud_resid[outliers_idx]

plot(predict(lm.quadratic), stud_resid, pch = 19, col = "darkgreen",
     main = "Studentized Residuals vs Fitted Values",
     xlab = "Fitted Values", ylab = "Studentized Residuals",
     ylim = c(-4, 4))
abline(h = c(-3, 0, 3), col = "red", lwd = 2, lty = 2)
```

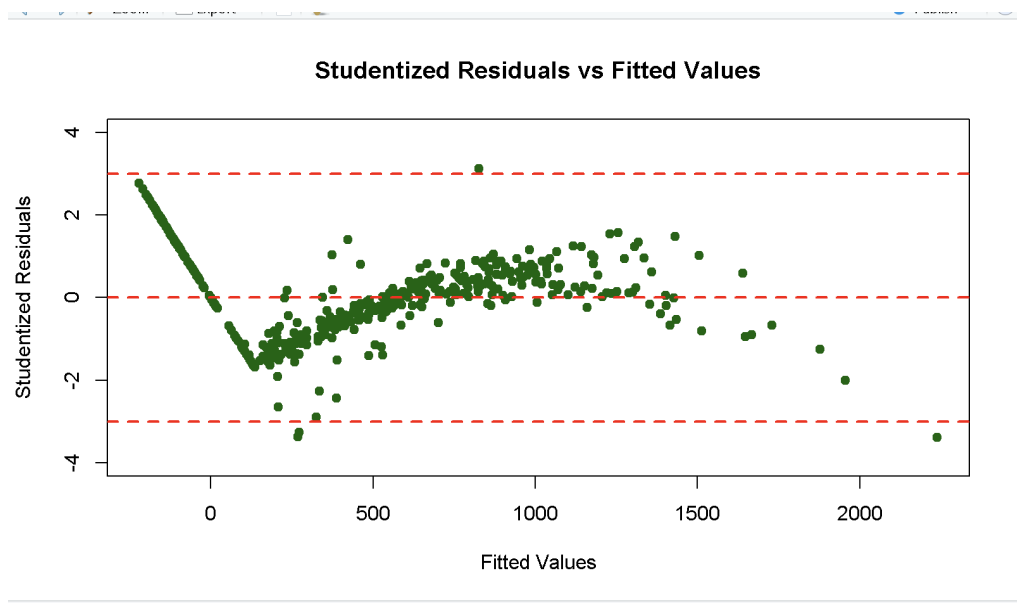


Figure 10: Studentized residuals vs fitted values (thresholds at ± 3)

Table 10: Identified Outliers

Row	Income	Limit	Age	Student	Balance	Stud.Res.
243	20.92	1233	47	Yes	16	-3.26
262	180.38	9310	67	Yes	1050	+3.12
321	16.28	1160	78	Yes	5	-3.37
324	182.73	13913	98	No	1999	-3.38

Inferences:

- 4 outliers out of 400 observations (1%) — well within normal expectations.

- Three of the four (243, 262, 321) are students, suggesting the uniform Student premium of \$429 does not fit all students equally well.
- Row 324 is the extreme boundary case — highest Limit, oldest age, maximum Balance.
- As noted in Lecture-29 slide 15, these appear to be legitimate data points rather than recording errors, so we retain them in the model.

7.7 High-Leverage Points

Per Lecture-29 slide 17, a point is considered high leverage if $h_{ii} > 2p/n$. Here $p = 8$ (including intercept) and $n = 400$, giving a threshold of 0.04.

R Code:

```
hat_vals = hatvalues(lm.quadratic)
n         = nrow(Credit)
p         = length(coef(lm.quadratic))
threshold = 2 * p / n

high_leverage_idx = which(hat_vals > threshold)
length(high_leverage_idx)
which.max(hat_vals)

plot(hat_vals, pch = 19, col = "darkgreen",
     main = "Leverage (Hat Values) by Observation",
     xlab = "Observation Index", ylab = "Leverage (h_ii)")
abline(h = threshold, col = "red", lwd = 2, lty = 2)
legend("topright",
      legend = "2p/n threshold (Lecture-29 Slide 17)",
      col = "red", lty = 2, lwd = 2)
```

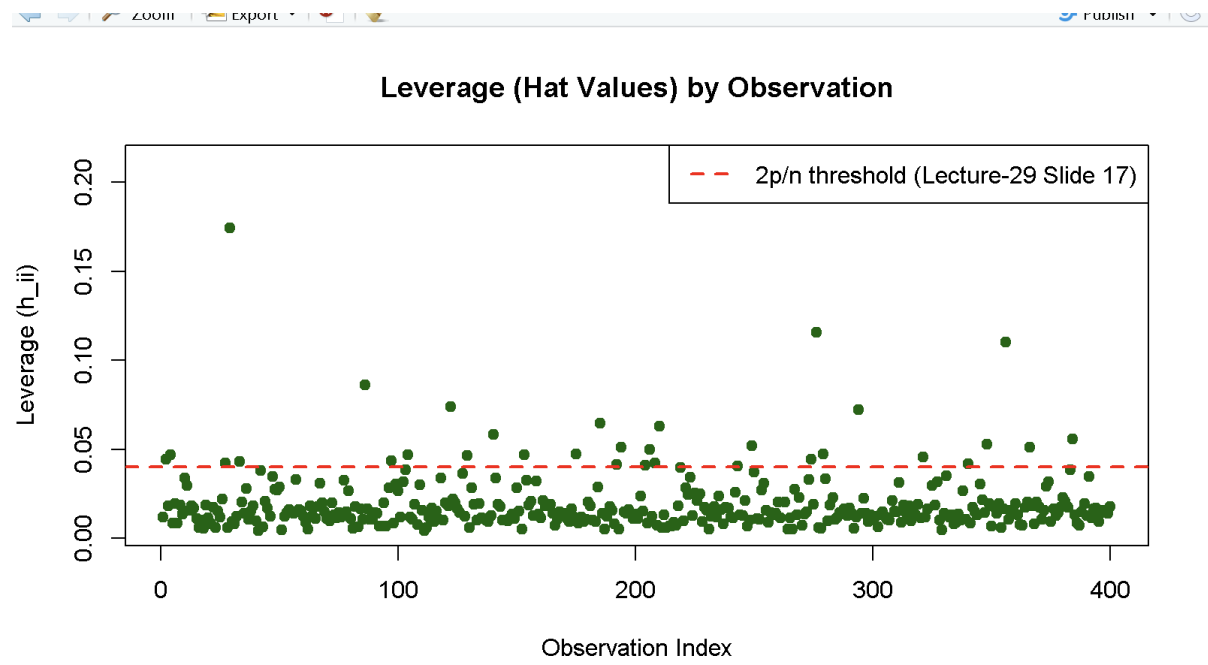


Figure 11: Leverage values by observation

- **34 high-leverage points** identified ($h_{ii} > 0.04$).
- **Maximum leverage:** row 324 ($h_{ii} = 0.2126$) — oldest person (age 98), highest credit limit, highest balance.
- **Critical finding:** all 4 outliers (243, 262, 321, 324) are also high-leverage points.

Inference: As warned in Lecture-29 slide 16, high-leverage points affect the regression line more strongly than ordinary outliers. In our case, all four outliers are also high-leverage, making them the most concerning observations. However, these represent genuine but unusual individuals (e.g., a 98-year-old with the highest credit limit) rather than data errors, so we retain them in the model.

7.8 Collinearity (Re-Check on Final Model)

R Code:

```
vif(lm.quadratic)
```

Table 11: VIFs for the Final Model

Variable	VIF
Income	2.76
Limit	2.70
Cards	1.00
Age	1.04
Student	1.00

Inference: Per Lecture-29 slide 20, a VIF above 5 or 10 indicates problematic collinearity. All VIFs in our final model are well below 5, confirming that collinearity is no longer an issue.

8 Summary of Findings

8.1 Final Model

$$\begin{aligned}\widehat{\text{Balance}} = & -283.9 - 6.44\text{Income} - 0.0205\text{Income}^2 \\ & + 0.170\text{Limit} + 1.03 \times 10^{-5}\text{Limit}^2 \\ & + 22.41\text{Cards} - 0.738\text{Age} + 440.9\text{Student}\end{aligned}$$

8.2 Model Progression

Table 12: Comparison of All Models Built

Model	R^2	Adj. R^2	RSE
Simple (Income only)	0.215	0.213	407.9
Multiple (6 quant, with Rating)	0.878	0.876	161.6
Multiple (Quant + Qual, no Rating)	0.9545	0.9533	99.34
Backward Selection (5 variables)	0.9541	0.9535	99.12
Quadratic Final Model	0.9695	0.969	81.01

8.3 Six Problems — Summary

Table 13: Check of All Six Regression Assumptions (Lecture-29)

#	Problem	Method Used	Result
1	Non-linearity	Residuals vs Fitted	Found, partly remedied
2	Correlated errors	Durbin-Watson	No problem ($p = 0.606$)
3	Non-constant variance	Residuals vs Fitted	Confirmed (curved pattern)
4	Outliers	Studentized residuals	4 outliers (1%)
5	High leverage	Hat values ($h > 2p/n$)	34 points; all 4 outliers
6	Collinearity	VIF	Fixed after dropping Rating

8.4 Conclusion

The final quadratic model explains **96.95%** of the variance in credit card balance with an average error of about \$81. The five main drivers of credit card balance are *Credit Limit*, *Student Status*, *Income*, *Number of Cards*, and *Age*. Qualitative variables like Gender, Married, and Ethnicity do not contribute significantly. Some model limitations remain — notably heteroscedasticity and a cluster of zero-balance observations that a linear model cannot fully capture — but addressing these would require more advanced techniques beyond the scope of this course.

References

1. James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). *An Introduction to Statistical Learning with Applications in R*. Springer.
2. Montgomery, D. C., Peck, E. A., and Vining, G. G. *Introduction to Linear Regression Analysis*.
3. Lecture notes from Statistical Methods (MA 2220), Dr. T. V. Rao, Mahindra University.